Solution Manual for The Art of Computer Systems Performance Analysis

Techniques for Experimental Design, Measurement, Simulation, and Modeling

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1.1

- Compare the ratio with system A as the base

System	Workload 1	Workload 2	Average
A	1	1	1
B	0.33	3	1.66

Considering the ratio of performance with system A as base, we conclude that system B is better.

- Compare the ratio with system B as the base

System	Workload 1	Workload 2	Average
A	3	0.33	1.66
B	1	1	1

Considering the ratio of performance with system B as base, we conclude that system A is better.

2.1 incomplete

2.2 Can be done

3.1 a. Measurements. Run your favourite programs and pick the one that runs them faster.

b. Use measurements and simulations of various network configuirations.

c. Measurement.

d. (a) Analytical modelling

(b) Analytical modelling and simulations.

(c) Extensive simulations and modelling.

3.2 a.
- Response time for commonly used programs.
- Failure rate (rate of crashing).
- Storage capacity.
- User-friendliness.

b.
- Query response time.
- Failure rate.
- Storage capacity.
- Usability.

c.
- Capacity.
- Response time.
- Failure rate.

d.
- Response time.

4.1 The following information is from SPEC (Standard Performance Evaluation Corporation) home page.

CPU benchmarks

CINT92, current release: Rel. 1.1. Integer benchmarks contains

Name	Application
espresso	Logic Design
li	Interpreter
eqntott	Logic Design
compress	Data Compression
sc	Spreadsheet
gcc	Compiler

CFP92, current release: Rel. 1.1.

Floating point benchmark suite contains

Name	Application
spice2g6	Circuit Design
doduc	Simulation
mdljdp2	Quantum Chemistry
wave5	Electromagnetism
tomcatv	Geometric Translation
ora	Optics
alvinn	Robotics
ear	Medical Simulation
mdljsp2	Quantum Chemistry
swm256	Simulation
su2cor	Quantum Physics
hydro2d	Astrophysics
nasa7	NASA Kernels
fpppp	Quantum Chemistry

More information about these benchmarks can be found in *http://performance.netlib.org/performa* web page.

4.2 A C program to implement sieve workload.

```
/*
 * seive.c : Program to implement sieve workload
 *
 */
```

```
#include <stdio.h>

#define MaxNum 8191             /* List all primes upto MaxNum */
#define NumIterations 10        /* Repeats procedure NumIterations times */

#define TRUE 1
#define FALSE 0

void main(void)
{
     int IsPrime[MaxNum+1];
     int i,k,Iteration;         /* Loop indexes */
     int NumPrimes;             /* Number of primes found */

     printf("Using Eratosthenes Sieve to find primes up to %d\n", MaxNum);
     printf("Repeating it %d times.\n",NumIterations);

     for (Iteration = 1; Iteration <= NumIterations; Iteration++)
     {
          /* Initialize all numbers to be prime */
          for (i = 1; i <= MaxNum; i++)
               IsPrime[i] = TRUE;

          i = 2;
          while (i*i <= MaxNum)
          {
               if (IsPrime[i])
               {
                    /* Mark all multiples of i to be nonprime */
                    k = i + i;
                    while (k <= MaxNum)
                         {
                              IsPrime[k] = FALSE;
                              k = k + i;
                              } /* of while  k */
               } /* of if IsPrime */
               i = i + 1;
```

```
          } /* of WHILE i*i */
          NumPrimes = 0;
          for (i = 1; i <= MaxNum; i++)
               /* Count the number of primes */
               if (IsPrime[i])
                    NumPrimes = NumPrimes + 1;
          printf("%d  primes\n",NumPrimes);

     } /* of for Iterations */
     /* The following can be added during debugging to list primes. */
     /* for (i = 0; i < MaxNum; i++)
        if (IsPrime[i]) printf("%d\n",i); */
}
```

The result of running the program

```
Using Eratosthenes Sieve to find primes up to 8191
Repeating it 10 times.
1029  primes
1029  primes
1029  primes
1029  primes
1029  primes
1029  primes
1029  primes
1029  primes
1029  primes
1029  primes
```

5.1 a. Cannot compare systems offering different services.

b.
- Metric: response time.
- Workload: Favourite programs: Word processor, spreadsheet.

c. Metric: response time, functionality Workload: A synthetic program which tests the versions using various operating system commands, operating system services.

d.
- Metric: Response time, reliability, time between failures
- Workload: A synthetic program generating representative floppy drive I/O requests

e.
- Metric: size of code, structure of code, execution time
- Workload: A representative set of programs in C and Pascal.

6.1 a.

$$\bar{t}_{\text{CPU}} = \frac{1}{n}\sum_{i=1}^{n} t_{CPU} = \frac{52}{7} = 7.428$$

$$\bar{n}_{\text{I/O}} = \frac{1}{n}\sum_{i=1}^{n} n_{I/O} = \frac{3,178}{7} = 454.0$$

$$\begin{aligned} s_{x_s}^2 &= \frac{1}{n-1}\sum_{i=1}^{n}(x_{si} - \bar{x}_s)^2 \\ &= \frac{1}{n-1}\left[\left(\sum_{i=1}^{n} x_{si}^2\right) - n\bar{x}_s^2\right] \\ &= \frac{518 - 7 \times 7.428^2}{6} = 4.69^2 \end{aligned}$$

Similarly,

$$s_{x_r}^2 = \frac{7,555,206 - 7 \times 454^2}{6} = 1009.3^2$$

b. *Normalize the variables to zero mean and unit standard deviation.* The normalized values x_s' and x_r' are given by

$$x_s' = \frac{x_s - \bar{x}_s}{s_{x_s}} = \frac{x_s - 7.428}{4.69}$$

$$x_r' = \frac{x_r - \bar{x}_r}{s_{x_r}} = \frac{x_r - 454}{1009.3}$$

The normalized values are shown in the fourth and fifth columns of Table b.

The other steps are similar to example 6.1.

Observation	Variables		Normalized Variables		Principal Factors	
No.	x_s	x_r	x'_s	x'_r	y_1	y_2
1	14	2735	1.401	2.260	2.589	−0.607
2	13	253	1.188	−0.199	0.699	0.981
3	8	27	0.122	−0.423	−0.213	0.385
4	6	27	−0.304	−0.423	−0.515	0.084
5	6	12	−0.304	−0.438	−0.525	0.094
6	4	91	−0.731	−0.360	−0.771	−0.262
7	1	33	−1.375	−0.450	−1.264	−0.674
$\sum x$	52	3,178	0.000	0.000	0.000	0.000
$\sum x^2$	518	7,555,206	6.000	6.000	9.970	2.017
Mean	7.42	454.0	0.000	0.000	0.000	0.000
Standard Deviation	4.69	1009.3	1.000	1.000	1.662	0.336

The correlation between CPU time (t_{CPU}) and number of I/O's ($n_{\text{I/O}}$) is 0.663. The principal factors y_1 and y_2 are:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{t_{\text{CPU}} - 7.43}{4.69} \\ \frac{n_{\text{I/O}} - 454.0}{1009.3} \end{bmatrix}$$

The first factor explains 1.661/(1.661+0.336) or 83% of total variation.

6.2 There is no unique solution to this exercise. Depending upon the choice of outliers, scaling technique, or distance metric, different results are possible, all of which could be considered correct. One solution using no outliers, range normalization to (0,1), and Euclidean distance starts with the the normalized values shown in the following:

Program	CPU time	I/O's
TKB	1.00	1.00
MAC	0.92	0.09
COBOL	0.54	0.01
BASIC	0.38	0.01
Pascal	0.38	0.00
EDT	0.23	0.03
SOS	0.00	0.01

BASIC, Pascal, EDT, COBOL, SOS, MAC, and TKB join the dendrogram at distances of 0.01, 0.15, 0.21, 0.38, 0.63, and 1.14, respectively.

Other possibilities are to discard TKB and MAC as outliers, normalize using the mean and standard deviation, and transform I/O's to a logarithmic scale. All these and similar alternatives should be considered correct provided a justification is given for the choice.

7.1 a. Hardware monitor as software monitor cannot measure time.

b. Software monitor becuase with hardware it is difficult to monitor software events.

c. Software monitor. Program reference is a software event

d. Hardware monitor. Virtual memory reference is a hardware event.

e. Hardware monitor. Software interferes with time measurements.

f. Software monitor. Database query is high-level (software) event.

7.2 Let us choose a network card our computer subsytem. Then the quantities that can be monitored using the different monitors are as follows

a. Software monitor. Total number of packets received, total number of packets sent. Number of error packets. Total bytes sent.

b. Hardware monitor. Record of all traffic to the card (using promiscuous mode).

c. Firmware monitor. The card be programmed to monitor traffic only from a particular node.

For the software monitor, using the quantities one could measure the average packet size, error rate in packets. For the hardware monitor, the record of traffic can be used to measure time between packet arrivals. For the firmware monitor, the number of packets received from a particular node can be measured.

8.1 a. Those with the largest number of terminal reads/writes per CPU second.

a. Find the average number of disk reads/writes per second of program X, and the maximum rate that the disk can support. The ratio gives you the number of copies of program X that can run simultaneously on the disk drive.

a. Find the mode of typical data recorded by the log and compare that with data of the benchmark. If they are close, then the benchmark is representative.

a. I/O bound programs - those with high "disk I/O's per CPU second" should be chosen for I/O optimization.

9.1 incomplete

9.2 incomplete

10.1 a. Bar chart, as intermediate values have no meaning.

b. Line chart, as intermediate values have meaning.

c. Bar chart, no meaning for intermediate value.

d. Line chart, intermediate values have meaning.

10.2 a. (a) Axes lables are not self-explanatory.
(b) Scales and divisions are not shown
(c) Curves are not labelled.

b. (a) Y-axis is not labelled.
(b) No Y-axis scales and division are not shown.
(c) Y-axis minimum and maximum are not appropriate.

c. (a) Scales and divisions are not shown.
(b) Too many curves.
(c) Curves are not individually labelled.

d. (a) Order of bars is wrong.
(b) Y-axis scales divisions not shown

10.3 incomplete

10.4 FOM = 73

10.5 FOM = 73

11.1 Raw Execution Time is a LB (Lower is better) parameter.

- Compare the ratio with system A as the base

Benchmark	System A	System B	System
I	1.00	2.00	3.00
J	1.00	1.50	0.50
K	1.00	0.33	0.67
Average	1.00	1.28	1.39

  Considering the ratio of performance with system A as base, we conclude that system A is better.

- Compare the ratio with system A as the base

Benchmark	System A	System B	System
I	0.50	1.00	1.50
J	0.67	1.00	0.33
K	3.00	1.00	2.00
Average	1.39	1.00	1.28

  Considering the ratio of performance with system B as base, we conclude that system B is better.

- Compare the ratio with system A as the base

Benchmark	System A	System B	System
I	0.33	0.67	1.00
J	2.00	3.00	1.00
K	1.50	0.50	1.00
Average	1.28	1.39	1.00

  Considering the ratio of performance with system C as base, we conclude that system C is better.

System A			
Test	Total	Pass	% Pass
1	a	ax	$100 \times x$
2	b	by	$100 \times y$
Total	$a+b$	$ax+by$	$\frac{100(ax+by)}{a+b}$

System B			
Test	Total	Pass	% Pass
1	c	cu	$100 \times u$
2	d	dv	$100 \times v$
Total	$c+d$	$cu+dv$	$100\frac{100(cu+dv)}{c+d}$

11.2 Consider two systems A and B with two experiments.

System A is better than B based on the individual experiments, if the following conditions are satisfied

$$100 \times x > 100 \times u$$

and

$$100 \times y > 100 \times v$$

These conditions simplify to

$$x > u \text{ and } y > v$$

System A is better than B based on total (both the experiments), if

$$\frac{100(ax+by)}{a+b} < \frac{100(cu+dv)}{c+d}$$

which simplifies to

$$\frac{(ax+by)}{a+b} < \frac{(cu+dv)}{c+d}$$

If any of the above conditions are satisfied then the percentages can be used for system A's advantage.

12.1 a. The servers are chosen independently with equal probablility, therefore the probability that server A is chosen $P(A) = \frac{1}{3}$.

b. $P(AorB) = P(A) + P(B) - P(AandB)$. Only one server at a time is selected, so $P(AandB) = 0$. Thus $P(AorB) = \frac{2}{3}$.

c. 0. Only one server at a time is selected.

d. $P(\bar{A}) = 1 - P(A) = \frac{2}{3}$.

e. Successive selections are independent, so we can multiple their probabilities. Thus $P(AA) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$.

f. All nine events are independent, each with probablility $\frac{1}{3}$, therefore the probability that they occur in sequence is $\frac{1}{3^9}$.

12.2 The distribution is geometric. The mean of a geometric distribution with pmf $(1-p)^{x-1}p$ is $\mu = \frac{1}{p}$. The variance $\sigma^2 = \frac{1-p}{p^2}$. The standard deviation $\sigma = \frac{\sqrt{1-p}}{p}$. The Coefficient of variation $COV = \frac{\sigma}{\mu} = \sqrt{1-p}$.

12.3 The mean of a Poisson distribution with pmf $\lambda^x \frac{e^{-\lambda x}}{x!}$ is $\mu = \lambda$. The variance $\sigma^2 = \lambda$. The $COV = \frac{\sigma}{\mu} = \sqrt{\lambda}$.

12.4 From 12.3, we know that the mean and variance of a Poisson distribution with pmf $(1-p)^{x-1}p$ are equal to λ. x and y are independent random variables.

a) $Mean(x+y) = Mean(x) + Mean(y) = 2\lambda$.

b) $Var(x+y) = Var(x) + Var(y) = 2\lambda$.

c) $Mean(x-y) = Mean(x) - Mean(y) = 0$.

d) $Var(x-y) = Var(x) + Var(y) = 2\lambda$.

e) $Mean(3x-4y) = 3Mean(x) - 4Mean(y) = -\lambda$.

f) $Var(3x-4y) = 9Var(x) + 16Var(y) = 25\lambda$. $COV = \sqrt{Var}/Mean = 5\sqrt{\lambda}$.

12.5

$$\begin{aligned} pdf = f(x) &= \frac{dF(x)}{dx} \\ &= \frac{e^{-x/a}}{a}\left[\sum_{i=0}^{m-1}\frac{(x/a)^i}{i!}\right] - e^{-x/a}\left[\sum_{i=0}^{m-2}\frac{(x/a)^i}{a*i!}\right] \\ &= \frac{\mathbf{x^{m-1}e^{-x/a}}}{\mathbf{(m-1)!a^m}} \end{aligned}$$

$$\begin{aligned} \text{Mean} = \mu &= \int_0^\infty x f(x)dx \\ &= \int_0^\infty \frac{x^m e^{-x/a}}{(m-1)!a^m}dx \\ &= \frac{1}{(m-1)!a^m}\int_0^\infty x^m e^{-x/a}dx \end{aligned}$$

Integrating by parts

$$\begin{aligned} &= \frac{1}{(m-1)!a^m}\left[-ax^m e^{-x/a}\right]_0^\infty + \frac{am}{(m-1)!a^m}\int_0^\infty x^{m-1}e^{-x/a}dx \\ &= \frac{am}{(m-1)!a^m}\int_0^\infty x^{m-1}e^{-x/a}dx \\ &= \frac{a^m m!}{(m-1)!a^m}\int_0^\infty e^{-x/a}dx \\ &= m\int_0^\infty e^{-x/a}dx \\ &= am\left[-e^{-x/a}\right]_0^\infty \\ &= am[0-(-1)] \\ &= \mathbf{am} \end{aligned}$$

$$\begin{aligned} \text{Variance} = \sigma^2 &= \int_0^\infty (x-\mu)^2 f(x)dx \\ &= \frac{1}{(m-1)!a^m}\int_0^\infty (x-am)^2 x^{m-1}e^{-x/a}dx \\ &= \frac{1}{(m-1)!a^m}\int_0^\infty (x^{m+1} - 2amx^m + a^2m^2x^{m-1})e^{-x/a}dx \end{aligned}$$

$$\begin{aligned} &= \frac{a^2(m+1)m - 2a^2m^2 + a^2m^2}{(m-1)!a^m}\int_0^\infty x^{m-1}e^{-x/a}dx \\ &= \mathbf{a^2m} \end{aligned}$$

Mode is the maximum possible probability.

f(x) is maxium when $\frac{df(x)}{dx} = 0$

$$\begin{aligned} \frac{df(x)}{dx} = \frac{1}{(m-1)!a^m}((m-1)x^{m-2}e{-}x/a - x^{m-1}e^{-x/a}) &= 0 \\ \frac{x^{m-1}e^{-x/a}}{(m-1)!a^m}((m-1) - x/a) &= 0 \\ \text{Therefore mode occurs at } x &= \mathbf{a(m-1)} \end{aligned}$$

$$\text{C.O.V} = \frac{\sigma}{\mu} = \frac{\sqrt{a^2m}}{am} = \frac{1}{\sqrt{m}}$$

12.6

$$\begin{aligned} pdf = f(x) &= \frac{dF(x)}{dx} \\ &= ax^{-(a+1)} \\ \text{Mean} = \mu &= \int_1^\infty xf(x)dx \\ &= \int_1^\infty ax^{-a}dx \\ &= a\left[\frac{x^{-a+1}}{(-a+1)}\right]_1^\infty \\ &= \frac{\mathbf{a}}{\mathbf{a-1}} \\ \text{Variance} = \sigma^2 &= \int_1^\infty (x-\mu)^2 f(x)dx \\ &= \int_1^\infty (x - \frac{a}{a-1})^2 ax^{-(a+1)}dx \end{aligned}$$

Integrating by parts

$$
\begin{aligned}
&= a\left[(x-\frac{a}{a-1})^2\frac{x^{-a}}{-a}\right]_1^\infty - 2\int_1^\infty (x-\frac{a}{a-1})x^{-a}dx \\
&= \frac{1}{(a-1)^2} - 2\left[(x-\frac{a}{a-1})\frac{x^{-a+1}}{-a+1}\right]_1^\infty + \frac{2}{-a+1}\int_1^\infty x^{-a+1}dx \\
&= \frac{1}{(a-1)^2} - \frac{2}{(a-1)^2} + \frac{2}{-a+1}\left[\frac{x^{-a+2}}{-a+2}\right]_1^\infty \\
&= \frac{-1}{(a-1)^2} + \frac{2}{(a-1)(a-2)} \\
&= \frac{2a-2-a+2}{(a-1)^2(a-2)} \\
&= \frac{\mathbf{a}}{\mathbf{(a-1)^2(a-2)}} \\
\text{C.O.V} = \frac{\sigma}{\mu} &= \sqrt{\frac{a}{(a-1)^2(a-2)}}\frac{a-1}{a} \\
&= \frac{1}{\sqrt{a(a-2)}} \\
&= [\mathbf{a(a-2)}]^{\mathbf{\frac{1}{2}}}
\end{aligned}
$$

12.7 The pdf for normal distribution is given by.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Here $\mu = 5$ and $\sigma = 1$. Hence,

$$f(x) = \frac{1}{\sqrt{2\pi}}e^{\frac{-(x-5)^2}{2}}$$

For pdf values for $x = 1, 2 \ldots 8$ are tabluated below.

x	f(x)
1	0.000134
2	0.004431
3	0.053991
4	0.241971
5	0.398942
6	0.241971
7	0.053991
8	0.004431
Total	0.999862

a. $P(X > 8) = 1 - P(X \leq 8) = 1 - \sum_{i=1}^{8} f(i) = \mathbf{0.000138}$ (xxx answer in the book is wrong)

b. $P(X < 6) = \sum_{i=1}^{5} f(i) = \mathbf{0.6995}$ (xxx answer in the book is wrong)

c. $f(4) + f(5) + f(6) + f(7) = 0.936875 = \mathbf{93.68}\%$ (xxx answer in the book is wrong)

d. $P(x \leq \mu + z_\alpha \sigma) = \alpha$ Here $\alpha = 0.95$ From appendix table A.2, $z_{0.95} = 1.645$. Hence $x = 5 + 1.645 \times 1 = \mathbf{6.645}$ seconds

12.8 a. The distribution is not skewed, nor is the data categorical, so we use the Mean.

b. The total number of packets makes sense; also the distribution is not skewed, so we use the Mean.

c. The distribution is skewed, so we use the Median.

d. The keywords constitute categorical information, hence we use the Mode.

12.9 a. CPU type is a category, so we would use Mode to summarize it.

b. Memory size is typically skewed - most personal compters have approximately the same amount of memory, but a few users have lots of memory - so the Median is the best choice.

c. Disk type is a category, so we would use the Mode.

d. Number of peripherals is skewed, so the Median is a good choice.

e. Using the same logic as for memory size and number of peripherals, we choose the Median.

12.10 Since the ratio of maximum to minimum is very high, use the median. The geometric mean can also be used if a logarithmic transformation can be justified based on physical considerations.

12.11 Arithmetic mean since the data is very clustered together (not skewed) and $y_{\max}/y_{\min}$ ratio is small.

12.12 Use SIQR since the data is skewed.

12.13 Range = 9 to 39 Variance = 90 10-percentile = [1 + (32)(0.10)] = 4th element = 14 90-percentile = [1 + (32)(0.9)] = 30th element = 38 Semi-Interquartile Range(SIQR) = $\frac{Q_3 - Q_1}{2}$ Q_1 = 9th element = 21 Q_3 = 25th element = 34 SIQR = 7/2 a= 3.5 Coefficient of Variation = 0.35

Use the coefficient of variation (or standard deviation) since the data is not skewed.

12.14 The normal quantile-quantile plot for this data is shown in Figure 20.2 of the book. From the plot, the errors do appear to be normally distributed.

13.1 The normal distribution has the linearity property. Hence, the means get added, when sum of two normal distribution are taken. The variance is given by

$$\sigma = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

a. From central limit theorem $N(\mu, \sigma/\sqrt{n})$ is the distribution of the sample means. Here $\sigma = 1$. Hence the distribution is $\mathbf{N(\mu, 1/\sqrt{n})}$

b. $mean = 0$. $\sigma = \sqrt{1/n + 1/n} = \sqrt{2/n}$. Hence the distribution is $\mathbf{N(0, \sqrt{2/n})}$ (xxx answer in the book is wrong)

c. $mean = \mu + \mu = 2\mu$. $\sigma = \sqrt{1/n + 1/n} = \sqrt{2/n}$. Hence the distribution is $\mathbf{N(2\mu, \sqrt{2/n})}$.

d. $mean = 1/2(\mu + \mu) = \mu$. $\sigma = \sqrt{1/4n + 1/4n} = 1/\sqrt{2n}$. Hence the distribution is $\mathbf{N(\mu, 1/\sqrt{2n})}$.

e. The sum of square of normal variates has the chi-square $\chi^{\mathbf{2}}(\mathbf{n})$ distribution.

f. The sum of the variance has $\chi^{\mathbf{2}}(\mathbf{2n})$ distribution.

g. The ratio of two chi-square distribution has an F distribution. Here both numerator and denominator have the same chi-squre distribution. Hence, $F(n, n)$ is the distribution for the ratio of variances.

h. If x is normal variate and $y \sim \chi^2(\nu)$ then $x/\sqrt{y/\nu}$ where ν is the degrees of freedom, has a t distribution. Here $(x - \mu)$ is normal variate. s_x has $\chi^2(n)$ distribution. The degrees of freedom is n, hence $(x-\mu)/(s_x/\sqrt{n})$ has $t(n)$ distribution.

13.2 The numbers in the sorted order is { 9, 11, 13, 14, 15, 15, 16, 19, 21, 23, 23, 23, 23, 24, 24, 25, 28, 28, 29, 31, 33, 33, 34, 34, 34, 35, 35, 36, 36, 38, 39, 42, 45}. There are $n = 33$ numbers.

a. The $100p$-percentile is $x_p = x_{([1+(n-1)p])}$. Therefore $x_{[1+(33-1)0.1]} = x_4 = \mathbf{14}$ and $x_{[1+(33-1)0.9]} = x_{30} = \mathbf{38}$.

b. Mean $= 1/33 \sum_{i=1}^{33} x_i = 888/33 = \mathbf{26.91}$

c. $s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2 = 90.1477$. There $s = \sqrt{90.1477} = 9.4946$.

A 90% confidence interval for the mean $= 26.91 \mp (1.645)(9.4946)/\sqrt{33}$

$$= (\mathbf{24.19}, \mathbf{29.63})$$

d. Number of programs with less than or equal 25 I/O's = 19. Fraction $= 19/33 = \mathbf{0.485}$.

A 95% confidence interval for the fraction $= p \mp z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{n}}$

$$= 0.485 \mp z \sqrt{\frac{0.485(1-0.485)}{33}}$$
$$= 0.485 \mp (1.960)(0.087)$$
$$= (\mathbf{0.314}, \mathbf{0.656})$$

(xxx answer in book is wrong, gives 90% C.I's)

e. One-sided confidence interval for mean is given by $(\bar{x}, \bar{x} + z_{1-\alpha} s/\sqrt{n})$ or $(\bar{x} - z_{1-\alpha} s/\sqrt{n}, \bar{x})$.

$(26.91 - (1.282)(9.4946)/\sqrt{33}, 26.91)$ or $(26.91, 26.91 + (1.282)(9.4946)/\sqrt{33})$

$$(\mathbf{24.79}, \mathbf{26.91}) \text{ or } (\mathbf{26.91}, \mathbf{29.03})$$

13.3 The standard deviation for the codes are $s_{RISC-I} = 4952.17$, $s_{Z8002} = 4948.37$,
$s_{VAX-11/780} = 2951.43$, $s_{PDP-11/70} = 2442.98$, $s_{C/70} = 2460.11$. The 90% confidence interval is given by $\bar{x} \mp t_{[0.95;10]} s/\sqrt{11}$, since $n = 11$ is less than 30. The confidence intervals are $CI_{RISC-I} = (-345.56, 5065.56)$, $CI_{Z8002} = (-563.30, 4843.66)$, $CI_{VAX-11/780} = (-181.39, 3033.57)$, $CI_{PDP-11/70} = (-16.33, 2653.06)$, $CI_{C/70} = (-46.05, 2642.05)$. (xxx answer in the book is wrong)

Let us choose RISC-I and Z8002 as the two systems.

a. The confidence intervals for both the processors include 0, so if the processors are not different.

b. incomplete, (question not clear??)

14.1 incomplete (refer math book to use Langrange multiplier technique)

14.2 A linear model to predict disk I/O's as a function of CPU time can be developed as follows

For this data:

$n = 7$, $\Sigma xy = 3375$, $\Sigma x = 66$, $\Sigma x^2 = 828$, $\Sigma y = 271$, $\Sigma y^2 = 13,855$, $\bar{x} = 9.43$, $\bar{y} = 38.71$. Therefore,

$$\begin{aligned} b_1 &= \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n(\bar{x})^2} = \frac{3375 - 7 \times 9.43 \times 38.71}{828 - 7 \times (9.43)^2} = 3.9886 \\ b_0 &= \bar{y} - b_1\bar{x} = 38.71 - 3.9886 \times 9.43 = 1.0975 \end{aligned}$$

The desired model is

$$\text{Number of disk I/O's} = 1.0975 + 3.9886(\text{CPU time})$$

$$\text{SSE} = \Sigma y^2 - b_0\Sigma y - b_1\Sigma xy = 13,855 - 1.0975 \times 271 - 3.9886 \times 3375 = 126.05$$

$$\text{SST} = \text{SSY} - \text{SS0} = \Sigma y^2 - n(\bar{y})^2 = 13,855 - 7 \times (38.71)^2 = 3365.75$$

$$\text{SSR} = \text{SST} - \text{SSE} = 3365.75 - 126.05 = 3239.70$$

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{3239.70}{3365.75} = 0.9625$$

Thus, the regression explains 96% of CPU time's variation.

The mean squared error is:

$$\text{MSE} = \frac{\text{SSE}}{\text{Degrees of Freedom for Errors}} = \frac{126.05}{5} = 25.21$$

The standard deviation of errors is:

$$s_e = \sqrt{\text{MSE}} = \sqrt{25.21} = 5.021$$

$$s_{b_0} = s_e\left[\frac{1}{n} + \frac{\bar{x}^2}{\Sigma x^2 - n\bar{x}^2}\right]^{1/2} = 5.021\left[\frac{1}{7} + \frac{(9.43)^2}{828 - 7 \times 9.43 \times 9.43}\right]^{1/2} = 3.8091$$

$$s_{b_1} = \frac{s_e}{[\Sigma x^2 - n\bar{x}^2]^{1/2}} = \frac{5.021}{[828 - 7 \times 9.43 \times 9.43]^{1/2}} = 0.3502$$

The 90% confidence interval for b_0 is

$$1.0975 \mp (2.015)(3.8091) = 1.0975 \mp 7.6753 = (-6.5778,\ 8.7728)$$

Since this includes zero b_0 is not significant.
The 90% confidence interval for b_0 is

$$3.9886 \mp (2.015)(0.3502) = 3.9886 \mp 0.7057 = (3.2829,\ 4.6943)$$

a. Only b_1 is significant.

b. **96%** (xxx answer in book is wrong it said 97

c. $y_{expected} = 1.0975 + 3.9886 * 40 = \mathbf{160.6415}$

d.

$$s_{\hat{y}_{1p}} = 5.021 \left[1 + \frac{1}{7} + \frac{(40 - 9.43)^2}{828 - 7(9.43)^2}\right]^{1/2} = 11.9768$$

$$\begin{aligned} \text{The 90\% confidence interval for a single prediction} &= 160.6415 \mp (2.015)(11.9768) \\ &= 160.6415 \mp 24.1333 \\ &= (136.50,\ 184.77) \end{aligned}$$

(xxx answer in book is wrong (139.50, 181,60))

e.

$$s_{\hat{y}_{1p}} = 5.021 \left[\frac{1}{7} + \frac{(40 - 9.43)^2}{828 - 7(9.43)^2}\right]^{1/2} = 10.8735$$

$$\begin{aligned} \text{The 90\% confidence interval for predicted mean} &= 160.6415 \mp (2.015)(10.8735) \\ &= 160.6415 \mp 21.9101 \\ &= (138.73,\ 182.55) \end{aligned}$$

(xxx answer in book is wrong (141.45, 179.66))

14.3 For the data:
$n = 7$, $\Sigma xy = 16,388$, $\Sigma x = 1324$, $\Sigma x^2 = 326,686$, $\Sigma y = 66$, $\Sigma y^2 = 828$, $\bar{x} = 189.14$, $\bar{y} = 9.43$. Therefore,

$$\begin{aligned} b_1 &= \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n(\bar{x})^2} = \frac{16,388 - 7 \times 189.14 \times 9.43}{326,686 - 7 \times (189.14)^2} = 0.0512 \\ b_0 &= \bar{y} - b_1\bar{x} = 9.43 - 0.0512 \times 189.14 = -0.254 \end{aligned}$$

The desired model is
CPU time in milliseconds $= -0.254 + 0.0512 \times$ (memory size in kilobytes)

$SSE = 5.6984$, $SST = 205.5257$, $SSR = 199.8273$, $R^2 = 0.972$, $MSE = 1.1396$, $s_e = 1.0675$, $s_{b_0} = 0.8351$, $s_{b_1} = .00387$.

The 90% confidence intervals of b_0 and b_1 are $(-1.9367, 1.4287)$ and $(0.0434, 0.0590)$; the intercept is zero but the slope is significant.

14.4 Elasped time = 0.196 (number of days) + 0.511; the 90% confidence intervals for the regression coefficients are (0.36, 0.66) for the intercept and (0.18, 0.21) for the slope; both are significant. (Note: Calculations are similar to the solution for exercise 14.2)

14.5 Elapsed time = 0.074+0.009×(number of keys); $R^2 = 0.943$; the confidence intervals of the coefficients are (0.461,0.809) and (0.042,0.084), respectively. (Note: Calculations are similar to the solution for exercise 14.2)

14.6 Number of disk I/O's = 13.494 + 1.634 × (number of keys); $R^2 = 0.846$; the 90% confidence intervals for the coefficients are $(-35.627, 27.877)$ and (2.78, 10.47); b_0 is not significant. (Note: Calculations are similar to the solution for exercise 14.2)

14.7 Time = −15,315.96 + 49.557 (record size); $R^2 = 0.744$. Both parameters are significant. However, the scatter plot of the data shows a nonlinear relationship. The residuals versus predicted estimates show that the errors have a parabolic trend. This indicates that the errors are not independent of the predictor variables and so either other predictor variables or some nonlinear terms of current predictors need to be included in the model.

15.1 (yyy question not clear, no x_5, but solution talks about x_5!?)

a. $R = 0.95$ $R^2 = 0.9025$. So 90.25% of variance is explained by the regression.

b. Yes

c. x_5

d. x_1

e. x_2, x_3, and x_4

f. Multicollinearity possible

g. Compute correlation among predictors and reduce the number of predictors.

Table 15.1: Time to Encrypt a k-bit Record (after log tranformation)

15.2

$log_2(k)$	Uniprocessor	Multiprocessor
2.107	1.968	1.826
2.408	2.679	2.550
2.709	3.532	3.371
3.010	4.405	4.231

Let us use the following model:

$$y = b_0 + b_1 x_1 + b_2 x_2$$

where y is log(time), x_1 is the key size and x_2 is a binary variable. $x_2 = 1 \Rightarrow$ multiprocessor and $x_2 = 0 \Rightarrow$uniprocessor;

In this case:

$$\mathbf{X} = \begin{bmatrix} 1 & 2.107 & 0 \\ 1 & 2.408 & 0 \\ 1 & 2.709 & 0 \\ 1 & 3.010 & 0 \\ 1 & 2.107 & 1 \\ 1 & 2.408 & 1 \\ 1 & 2.709 & 1 \\ 1 & 3.010 & 1 \end{bmatrix}$$

$$\mathbf{X}^{\mathrm{T}}\mathbf{X} = \begin{bmatrix} 8 & 20.468 & 4 \\ 20.468 & 53.273 & 10.234 \\ 4 & 10.234 & 4 \end{bmatrix}$$

$$\mathbf{C} = (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1} = \begin{bmatrix} 7.475 & -2.824 & -0.25 \\ -2.824 & 1.104 & 0 \\ -0.25 & 0 & 0.5 \end{bmatrix}$$

$$\mathbf{X}^{\mathrm{T}}\mathbf{y} = \begin{bmatrix} 24.562 \\ 65.280 \\ 11.978 \end{bmatrix}$$

The regression parameters are:

$$\mathbf{b} = (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y} = (-3.739, 2.691, -0.152)^{\mathrm{T}}$$

The regression equation is:

$$\log(\text{time}) = -3.738 + 2.690 \log(\text{key size}) - 0.152x_2$$

$R^2 = 0.9981$. The confidence intervals of the coefficients are $(-4.01, -3.46)$, $(2.58, 2.80)$, and $(-0.22, -0.08)$.

16.1 Each of the factors have 3 levels.

a. A full factorial experiment is necessary if there is significant interaction among factors. Hence, number of experiments is $3 \times 3 \times 3 = \mathbf{27}$.

b. If there is no interaction among factors, then simple design can be used. The number of experiments is $1 + (3-1) + (3-1) + (3-1) = \mathbf{7}$ 9

c. A fractional factorial experiment can be used if the interaction is small among factor. Hence, number of experiments is $3^{3-1} = \mathbf{9}$

(xxx answer in book is wrong. The answers for (b) and (c) are interchanged)

17.1 The sign table for this data is given below

I	A	B	C	AB	AC	BC	ABC	y
1	−1	−1	−1	1	1	1	−1	100
1	1	−1	−1	−1	−1	1	1	120
1	−1	1	−1	−1	1	−1	1	40
1	1	1	−1	1	−1	−1	−1	40[1]
1	−1	−1	1	1	−1	−1	1	15
1	1	−1	1	−1	1	−1	−1	10
1	−1	1	1	−1	−1	1	−1	30
1	1	1	1	1	1	1	1	50
385	15	−105	−175	−15	15	215	65	Total
48.125	1.875	−13.125	−21.875	−1.875	1.875	26.875	8.125	Total/8

$$\begin{aligned}
\text{SST} &= 2^3(q_A^2 + q_B^2 + q_C^2 + q_{AB}^2 + q_{AC}^2 + q_{BC}^2 + q_{ABC}^2) \\
&= 8(1.875^2 + 13.125^2 + 21.875^2 + 1.875^2 + 1.875^2 + 26.875^2 + 8.125^2) \\
&= 28.125 + 1378.125 + 3570.125 + 28.125 + 28.125 + 5778.125 + 528.125 \\
&= 11597.025
\end{aligned}$$

a. $q_0 = 48.13$, $q_A = 1.88$, $q_B = -13.13$, $q_C = -21.88$, $q_{AB} = -1.88$, $q_{AC} = 1.88$, $q_{BC} = 26.88$, and $q_{ABC} = 8.13$

b. The portion of variation explained by the seven effects are: 28.125/11597.025 (0.24%),1378.125/11597.025 (11.8(33.01%), 28.125/11597.025 (0.24%), 28.125/11597.025 (0.24%), 5778.125/11597.025 (48.82%), and 528.125/11597.025 (4.55%), respectively.

c. Sorting according to their coefficient values, the factors with decreasing order of importance are : BC, C, B, ABC, A, AB, AC

18.1 Let A indicate workload and B indicate Processor's used.

I	A	B	AB	y	Mean $\bar{y}$
1	-1	-1	1	(41.16, 39.02, 42.56)	40.913
1	1	-1	-1	(51.50,52.50,50.50)	51.500
1	-1	1	-1	(63.17,59.25,64.23)	62.217
1	1	1	1	(48.08,48.98,47.10)	48.053
202.683	-3.577	17.857	-24.751	Total	
50.67	-0.894	4.464	-6.188	Total/4	

The effects are 50.67, 4.46, -0.89, and -6.19. The effect of workloads (-0.89) is not significant. Interactions explain 62.67% of the variation.

19.1 The following sign table with $I = ACD$ as the generator polynomial is used to analyze the 2^{4-1} design.

I	A	B	C	AB	D	BC	BD	y
1	−1	−1	−1	1	1	1	−1	40
1	1	−1	−1	−1	−1	1	1	100
1	−1	1	−1	−1	1	−1	1	20
1	1	1	−1	1	−1	−1	−1	120
1	−1	−1	1	1	−1	−1	1	15
1	1	−1	1	−1	1	−1	−1	30
1	−1	1	1	−1	−1	1	−1	10
1	1	1	1	1	1	1	1	50
385	215	15	−175	65	−105	15	−15	Total
48.125	26.875	1.875	−21.875	8.125	−13.125	1.875	−1.875	Total/8

a. $q_0 + q_{ACD} = 48.13$, $q_A + q_{CD} = 1.88$, $q_B + q_{ABCD} = -13.13$, $q_C + q_{AD} = -21.88$, $q_{AB} + q_{BCD} = -1.88$, $q_{AC} + q_D = 1.88$, $q_{BC} + q_{ABD} = 26.88$, and $q_{ABC} + q_{BD} = 8.13$

b. 0.24%, 11.88%, 33.01%, 0.24%, 0.24%, 49.82%, 4.55%

c. BC, C, B, BD, A, AB, D. Higher order interactions are assumed smaller.

d. The generator is $I = ACD$. The confoundings are

$$I = ACD,\ A = CD,\ B = ABCD,\ C = AD,\ D = AC,\ AB = BCD,$$
$$BC = ABD \text{ and } ABC = BD$$

e. $I = ABCD$ may be better since its resolution will be IV.

f. R_{III}, since the generator is $I = ACD$.

19.2 Yes, $I = ABC$ is a 2^{4-1}_{III} design; yes, $I = AB$ is a 2^{4-1}_{II} design; yes, $I = ABCD$ is a 2^{4-1}_{IV} design.

20.1 Rewrite the given equation as

$$\alpha_j = \sum_{i=1}^{r}\sum_{k=1}^{a} a_{ikj} y_{ik}$$

We know that

$$\alpha_j = \bar{y}_{.j} - \mu = \bar{y}_{.j} - \bar{y}_{..}$$

Expanding the terms for $\bar{y}_{.j}$ and $\bar{y}_{..}$ we get the following equation.

$$\alpha_j = \frac{1}{r}\sum_{i=1}^{r} y_{ij} - \frac{1}{ar}\sum_{i=1}^{r}\sum_{k=1}^{a} y_{ik}$$

Collecting the terms we get

$$\alpha_j = (\frac{1}{r} - \frac{1}{ar})\sum_{i=1}^{r} y_{ij} - \frac{1}{ar}\sum_{i=1, i\neq j}^{r}\sum_{k=1}^{a} y_{ik}$$

Comparing the coefficients of y_{ij} we get

$$a_{ikj} = \begin{cases} \frac{1}{r} - \frac{1}{ra}, & k = j \\ \frac{-1}{ra}, & \text{Otherwise} \end{cases}$$

(xxx answer in book is wrong, it gives $1/ar$ instead of $-1/ar$)

The variance of e_{ij} can be written as

$$\begin{aligned}
\sigma^2_{e_{\alpha_j}} &= \sigma^2_e \sum_{i=1}^{r}\sum_{k=1}^{a} a_{ikj} \\
\sigma^2_{e_{\alpha_j}} &= \left[r\left(\frac{1}{r} - \frac{1}{ra}\right)^2 + (ar - r)\left(\frac{1}{ra}\right)^2 \right] \sigma^2_e \\
&= \left[\frac{1}{ra^2}(a-1)^2 + \frac{1}{ra^2}(a-1) \right] \sigma^2_e \\
&= \left[\frac{(a-1)}{ra^2}((a-1)+1) \right] \sigma^2_e \\
&= \frac{(a-1)\sigma^2_e}{ar}
\end{aligned}$$

Table 21.2: Computation of Effects for the Scheme versus Spectrum Study

Workload	Scheme86	Spect125	Spect62.5	Row Sum	Row Mean	R Effe
Garbage Collection	39.97	99.06	56.24	195.27	65.09	49.0
Pattern Match	0.958	1.672	1.252	3.882	1.294	−14.7
Bignum Addition	0.01910	0.03175	0.01844	0.0693	0.0231	−16.0
Bignum Multiplication	0.256	0.423	0.236	0.915	0.305	−15.7
Fast Fourier Transform (1024)	10.21	20.28	10.14	40.63	13.543	−2.5
Column Sum	51.413	121.467	67.88644	240.766		
Column Mean	10.283	24.293	13.577		16.051	
Column effect	−5.768	8.242	−2.474			

21.1 The computation of effects for Scheme versus Spectrum study is given in table 21.2

$$
\begin{aligned}
\text{SSY} &= \sum_{ij} y_{ij}^2 = 15,197.347 \\
\text{SS0} &= ab\mu^2 = 3 \times 5 \times (16.051)^2 = 3,864.519 \\
\text{SSA} &= b\sum_j \alpha_j^2 = 5 \times [(-5.768)^2 + (8.242)^2 + (-2.474)^2] = 536.605 \\
\text{SSB} &= a\sum_i \beta_i^2 = 3 \times [(49.039)^2 + (-14.757)^2 + (-16.028)^2 + (-15.747)^2 + (-2.508)^2] \\
&= 9,401.242 \\
\text{SST} &= \text{SSY} - \text{SS0} = 15,197.347 - 3,864.519 = 11,332.828 \\
\text{SSE} &= \text{SST} - \text{SSA} - \text{SSB} = 11,332.828 - 536.605 - 9,401.242 = 1394.981
\end{aligned}
$$

The differences of the effects of different processors are $-5.768 - 8.242 = -14.01$, $-5.768 + 2.474 = -3.294$ and $8.242 + 2.474 = 10.716$.

$$\text{MSE} = \frac{\text{SSE}}{(a-1)(b-1)}$$

$$MSE = \frac{1394.981}{(5-1)(3-1)} = 139.498$$

$$s_e = \sqrt{\text{MSE}} = \sqrt{139.498} = 11.811$$

Table 21.3: Computation of Effects the Intel iAPX 432 Study

System No.	Workload 484	Sieve	Puzzle	Acker	Row Sum	Row Mean	Row Effect
1	0.146	2.398	3.973	3.663	10.180	2.545	−0.5140
2	0.204	2.342	4.076	3.892	10.514	2.626	−0.4304
3	0.146	2.413	4.062	3.993	10.615	2.654	−0.4053
4	0.672	2.869	4.569	3.892	12.003	3.001	−0.0583
5	0.724	810.0	4.511	4.060	12.204	3.051	−0.0079
6	0.763	2.982	4.512	4.091	12.348	3.087	0.0281
7	0.114	2.292	3.963	3.439	9.808	2.452	−0.6069
8	0.176	2.391	3.964	3.489	10.019	2.505	−0.5541
9	0.863	2.884	4.643	4.045	12.436	3.109	0.0499
10	1.544	3.505	5.544	5.414	16.008	4.002	0.9430
11	1.152	3.505	5.217	5.414	15.290	3.822	0.7634
12	1.206	3.505	5.255	5.414	15.382	3.8455	0.7865
Column Sum	7.718	33.996	54.290	50.809	146.813		
Column Mean	0.643	2.833	4.524	4.234		3.059	
Column Effect	−2.416	−0.226	1.465	1.175			

The standard deviation for the differences $= s_e/\sqrt{2} = 11.811/1.414 = 8.3516$

The confidence intervals for the differences are

$$14.01 \mp 1.86 * 8.3516 = 14.01 \mp 15.53404 = (-29.544, 1.524)$$

$$-3.294 \mp 1.86 * 8.3516 = -3.294 \mp 15.53404 = (-18.828, 12.240)$$

$$10.716 \mp 1.86 * 8.3516 = 10.716 \mp 15.53404 = (-4.818, 26.250)$$

The processor are not signficantly different. The plots of residul errors does not show any trend and the plot of normal quantile-quantile plot does appear linear. But since the y_{max}/y_{min} is large a multiplicative model should be used.

21.2 The table after the log transformation is shown in table 21.3.

The ANOVA for Scheme versus Spectrum study is given in table 21.4, which agrees with the table 21.19 given in the book.

Table 21.4: ANOVA Table for the Intel iAPX 432 Study

Component	Sum of Squares	%Variation	DF	Mean Square	F-Comp.	F-Table
y	576.639					
$y_{..}$	449.159					
$y - y_{..}$	127.625	100.0%	47			
Workload	112.979	88.5%	3	37.7	1158.5	2.3
System	12.991	10.2%	11	1.2	37.9	1.8
Errors	1.07	0.8%	33	0.03		

$$s_e = \sqrt{\text{MSE}} = \sqrt{0.03} = 0.18$$

21.3 After logarithmic transformation, the table for computing effects is shown in table 21.5.

The ANOVA table for RISC Code size study is given in table 21.6.

The confidence intervals for effect differences are shown in table 21.7.

a. 1.18% variation is explained by the processors

b. 96.08% variation is due to workloads.

c. Yes, several processor pairs are significantly different. at 90% confidence level.

21.4 After log transformation, the table for computing effects (including 68000 column) is given in table 21.8.

The ANOVA table including 68000 column is given in table 21.9.

The confidence intervals of effect differences for RISC code study including column 68000 is given in table 21.10.

a. 1.65% variation is explained by the processors.

b. 96.03% variation is due to workloads.

c. Yes, several processor pairs are significantly different at 90% confidence level.

Table 21.5: Computation of Effects for the RISC Code Size Study

	Processors					Row	Row	Row
Workload	RISC-I	Z8002	VAX-11/780	PDP-11/70	C/70	Sum	Mean	Effect
E-String Search	2.16	2.11	2.00	2.06	2.00	10.34	2.07	-0.60 [2]
F-Bit Test	2.08	2.26	2.16	2.23	2.08	10.80	2.16	-0.51
H-Linked List	2.25	2.15	2.32	2.48	2.15	11.34	2.27	-0.40
K-Bit Matrix	2.46	2.57	2.46	2.57	2.50	12.57	2.51	-0.15
I-Quick Sort	3.00	3.04	2.95	3.04	2.95	14.97	2.99	0.33
Ackermann(3,6)	2.16	2.48	1.86	1.93	1.93	10.36	2.07	-0.59
Recursive Qsort	3.44	3.14	3.14	3.22	3.22	16.14	3.23	0.57
Puzzle (Subscript)	3.45	3.15	3.15	3.15	3.22	16.11	3.22	0.56
Puzzle (Pointer)	2.88	2.78	2.65	2.58	2.58	13.46	2.69	0.03
SED (Batch Editor)	4.25	4.25	4.03	3.95	3.95	20.42	4.08	1.42
Towers Hanoi (18)	1.98	2.38	1.89	1.98	1.83	10.06	2.01	-0.65
Column Sum	30.09	30.30	38.60	29.17	28.41	146.57		
Column Mean	2.74	2.75	2.60	2.65	2.58		2.66	
Column Effect	0.07	0.09	-0.06	-0.01	-0.08			

Table 21.6: ANOVA Table for RISC Code Size Study

Component	Sum of Squares	%Variation	DF	Mean Square	F-Comp.	F-Table
y	413.07					
$y_{..}$	390.60					
$y - y_{..}$	22.47	100.0%	54			
Workload	0.26	1.18%	4	0.07	4.31	2.09
System	21.59	96.08%	10	2.16	140.45	1.76
Errors	0.61	2.74%	40	0.02		

$$s_e = \sqrt{\text{MSE}} = \sqrt{0.02} = 0.12$$

Table 21.7: Confidence Intervals of Effect Differences in the RISC Code Size Study

	RISC-I	Z8002	VAX-11/780	PDP-11/70	C/70
RISC-I		(−0.11,0.07)†	(0.05, 0.22)	(−0.01,0.17)†	(0.06, 0.24)
Z8002			(0.07, 0.24)	(0.01, 0.19)	(0.08, 0.26)
VAX-11/780				(−0.14,0.04)†	(−0.07, 0.11)†
PDP-11/70					(−0.02, 0.16)†

† ⇒ Not significant

Table 21.8: Computation of Effects for the RISC Code Size Study

	Processors						Row	Row	
Workload	RISC-I	68000	Z8002	11/780	11/70	C/70	Sum	Mean	E
E-String Search	2.16	2.06	2.11	2.00	2.06	2.00	12.40	2.07	−0
F-Bit Test	2.08	2.16	2.26	2.16	2.23	2.08	12.96	2.16	–
H-Linked List	2.25	2.09	2.15	2.32	2.48	2.15	13.43	2.24	–
K-Bit Matrix	2.46	2.50	2.57	2.46	2.57	2.50	15.07	2.51	–
I-Quick Sort	3.00	2.84	3.04	2.95	3.04	2.95	17.82	2.97	
Ackermann(3,6)	2.16	0.00	2.48	1.86	1.93	1.93	10.36	2.07	–
Recursive Qsort	3.44	0.00	3.14	3.14	3.22	3.22	16.14	3.23	
Puzzle (Subscript)	3.45	3.40	3.15	3.15	3.15	3.22	19.51	3.25	
Puzzle (Pointer)	2.88	0.00	2.78	2.65	2.58	2.58	13.46	2.69	
SED (Batch Editor)	4.25	0.00	4.25	4.03	3.95	3.95	20.42	4.08	
Towers Hanoi (18)	1.98	0.00	2.38	1.89	1.98	1.83	10.06	2.01	–
Column Sum	30.09	15.05	30.30	38.60	29.17	28.41	161.62		
Column Mean	2.74	2.51	2.75	2.60	2.65	2.58		2.65	
Column Effect	0.09	−0.14	0.10	−0.05	−0.00	−0.07			

Table 21.9: ANOVA Table for RISC Code Size Study

Component	Sum of Squares	%Variation	DF	Mean Square	F-Comp.	F-Table
y	452.24					
$y_{..}$	428.23					
$y - y_{..}$	24.01	100.0%	60			
Workload	0.40	1.65%	5	0.08	4.61	1.99
System	23.05	96.03%	10	2.31	133.89	1.75
Errors	0.77	3.23%	45	0.02		

$s_e = \sqrt{\text{MSE}} = \sqrt{0.02} = 0.13$

Table 21.10: Confidence Intervals of Effect Differences in the RISC Code Size Study

	6800	Z8002	11/780	11/70	C/70
RISC-I	(0.13,0.32)	(−0.11,0.07)†	(0.04, 0.23)	(−0.01,0.18)†	(0.06, 0.25)
68000		(−0.34, −0.15)	(−0.19, 0.00)†	(−0.24, −0.05)	(−0.17, 0.02)†
Z8002			(0.06, 0.25)	(0.01, 0.20)	(0.08, 0.27)
11/780				(−0.15,0.04)†	(−0.08, 0.11)†
11/70					(−0.02, 0.16)†

† ⇒ Not significant

Table 22.11: Computation of Effects for exercise 22.1

	A			Row	Row	Row
B	A1	A2	A3	Sum	Mean	Effect
B1	3200.0	5120.0	8900.0	51660.0	5740.0	−4028.7
B2	4700.0	9400.0	19740.0	101520.0	11280.0	1511.3
B3	3200.0	4160.0	7360.0	44160.0	4906.7	−4862.0
B4	5100.0	5610.0	22440.0	99450.0	11050.0	1281.3
B5	6800.0	12240.0	28560.0	142800.0	15866.7	6098.0
Column Sum	23000.0	36530.0	87000.0	146530.0		
Column Mean	4600.0	7306.0	17400.0		9768.7	
Column effect	−5168.7	−2462.7	7631.3			

Table 22.12: Interactions

B	A1	A2	A3
B1	2628.7	1842.7	−4471.3
B2	−1411.3	582.7	828.7
B3	3462.0	1716.0	−5178.0
B4	−781.3	−2977.3	3758.7
B5	−3898.0	−1164.0	5062.0

22.1 The table for computing effects is given in table 22.11. The interactions are given in table 22.12. The Analysis of Variance table is given table 22.13. The 90% confidence intervals for effects are given in table 22.14. The 90% confidence intervals for the interactions are given in table 22.15. The 90% confidence intervals for the effect differences are given in table 22.16.

a. Yes. All processors are significantly different from each other.

b. 16.8%

c. All effects and interactions are significant.

Table 22.13: ANOVA Table exercise 22.1

Component	Sum of Squares	%Variation	DF	Mean Square	F-Comp.	F-Table
y	6810978816					
$y_{..}$	4294208256					
$y - y_{..}$	2516770560	100.0%	44			
A	1365256448	54.2%	2	682628224	109924.0	2.5
B	728826816	29.0%	4	182206704	29340.9	2.1
Interactions	422501056	16.8%	8	52812632	8504.4	1.9
Errors	186300	0.0%	30	6210		

$s_e = \sqrt{\text{MSE}} = \sqrt{6210} = 78.80$

Table 22.14: Confidence Intervals for Effects

Parameter	Mean Effect	Std. Dev.	Confidence Interval
μ	9768.7	11.7	(9748.7, 9788.6)
A			
A1	−5168.7	16.6	(−5196.9, −5140.5)
A2	−2462.7	16.6	(−2490.9, −2434.5)
A3	7631.3	16.6	(7603.1, 7659.5)
B			
B1	−4028.7	23.5	(−4068.5, −3988.8)
B2	1511.3	23.5	(1471.5, 1551.2)
B3	−4862.7	23.5	(−4901.9, −4822.1)
B4	1281.3	23.5	(1241.5, 1321.2)
B5	6098.0	23.5	(6058.1, 6137.9)

Table 22.15: Interactions

B	A1	A2	A3
B1	(2572.6, 2685.1)	(1786.3, 1899.1)	(−4527.7, −4414.9)
B2	(−1467.7,−1354.9)	(526.3,639.1)	(727.3,885.1)
B3	(3405.6,3518.4)	(1659.6,1772.4)	(−5234.4,−5121.6)
B4	(−837.7,−724.9)	(−3033.7,−2920.9)	(3702.3,3815.1)
B5	(−3954.4,−3841.6)	(−1220.4,−1107.6)	(5005.6,5118.4)

Table 22.16: Interactions

A1	A2	A3
A1	(−2754.8, −2657.2)	(−12848.8, −12751.2)
A1		(−10142.8, −10045.2)

23.1 The experiment number 9 maximizes T_I (66.1), so T_I is high for $A = -1$, $B = -1$, and $E = -1$; The experiment number 3 maximizes T_B (21.0), so, T_B is high for $A = -1$, $B = 1$, and $D = -1$.

23.2 The throughputs are ranked according to decreasing value of T_I and it is seen that, T_I is high for $A = -1$, $B = -1$, and $E = -1$; Similarily when the throughputs are ranked according to decreasing value of T_B it is seen that, T_B is high for $A = -1$, $B = 1$, and $D = -1$.

24.1 a. $y(t) = t + 0.2$ Countinuous state, deterministic, dynamic, linear, and unstable

b. $y(t) = t^2$ Continuous state, deterministic, dynamic, nonlinear, and unstable

c. $y(t+1) = y(t) + \Delta$, Δ is not an integer. Discrete time, continuous state, deterministic, dynamic, linear, and unstable

d. $n(t+1) = 2n(t) + 3$ Discrete time, deterministic, dynamic, linear, and unstable

e. $y(t) = sin(wt)$ Continuous time, continuous state, dynamic, nonlinear, and stable

f. $\bar{y}(t+1) = \bar{y}(t) + \Delta$ Discrete time, continuous state, probabilistic , dynamic, linear, and unstable

24.2 a. Since the number of factors this is best modeled by a *Trace-driven simulation.*

b. The known distribution can be used to generate the events. Hence this is best modeled by *Discrete-event simulation.*

c. The value of π is independent of time. Hence *Monte Carlo simulation* is best suited to find the value of π.

24.3 The *unit time* approach is a time-advancing mechanism to adjust the simulation clock. In this approach the time is incremeneted in small intervals and checks are done at each increment to see if there are any events which have to be scheduled. This approach is generally not used, since unnecessary increments and checks are done during idle time.

25.1 a. This is expected when the system is underloaded. Make sure that the system is in underloaded region

b. This is quite common when the system is overloaded. Make sure that the system is in overloaded region

c. This is expected.

d. This is uncommon and would require validation.

e. This is rare and would require serious validation effort.

25.2 The transient interval using the truncation method is 1, since 4 is neither the maximum nor the minimum of the remaining observations. However, this is incorrect, since the actual transient interval seems to be 6.

26.1 This is multiplicative LCG. Hence the maximum Maximum period is $2^{l-2} = 2^{4-2} = 4$; a must be $8i \mp 3$, that is 5 or 11. The seed must be odd.

26.2 The values of 24^n mod 31 for $n = 1, \ldots, 30$ are 24, 18, 29, 14, 26, 4, 3, 10, 23, 25, 11, 16, 12, 9, 30, 7, 13, 2, 17, 5, 27, 28, 21, 8, 6, 20, 15, 19, 22, 1. The smallest n that results in 1 is 30. Yes, 24 is a primitive root of 31.

26.3 2, 6, 7, 8

26.4 1155

26.5 x_{10000} = 1,919,456,777

26.6

$$q = m \text{ div } a = 31 \text{ div } 11 = 2$$

$$r = m \text{ mod } a = 31 \text{ mod } 11 = 9$$

No, the seqence generated with and without Schrage method are different. Since, $q = 2$ and $r = 9$ do not satisfy the condition r less than q.

26.7

$$q = m \text{ div } a = 31 \text{ div } 24 = 1$$

$$r = m \text{ mod } a = 31 \text{ mod } 24 = 7$$

Since, $q = 1$ and $r = 7$ do not satisfy the condition $r < q$, this cannot be implemented using Schrage's method.

26.8 a. Primitive

b. Primitive

c. Not primitive. Since $(1 + x^2 + x^4)(1 - x^2) = 1 - x^6$.

d. Primitive

26.9 a. 21.

b. 9. Since $(x^3 - 1)(x^6 + x + 1) = x^9 - 1$

Table 26.17: Tausworthe method

Step1: Copy seed Y_1	111111	100000	110000	101000	111100
Step2: 1-bit Right shift Y_2	011111	010000	011000	010100	011110
Step3: Xor $Y_3 = Y_1 \oplus Y_2$	100000	110000	101000	111100	100010
Step4: 5-bit Left shift Y_4	000000	000000	000000	000000	000000
Step5: Xor, $Y_5 = Y_3 \oplus Y_4$	100000	110000	101000	111100	100010

c. 63.

d. 45. Since $(x^30 + x^15 + 1)(x^15 - 1) = x^45 - 1$

26.10 The characteristic polynomial is $x^6 + x + 1$. In this case $r = 1$, $q = 6$, and $q - r = 5$. We need a 1-bit right shift and 5-bit left shift. The initial seed is X = 111111.

The sequence of calculations are show in table 26.17. Hence the first five 6-bit numbers are 0.000001_2, 0.000011_2, 0.000101_2, 0.001111_2, 0.010001_2.

26.11 In both cases, the additive parameter c should be replaced by $c \mod m$.

26.12 The first 20 numbers and their binary representations are listed in table 26.18. From the table it can be seen that the period of the lth bit is 2^l.

Table 26.18: Random Numbers Generated by the LCG: $x_n = 13x_{n-1} + 11 \bmod 2^{16}$

n	x_n	
	Decimal	Binary
1	24	00000000 00011000
2	323	00000001 01000011
3	4,210	00010000 01110010
4	54,741	11010101 11010101
5	56,284	11011011 11011100
6	10,807	00101010 00110111
7	9,430	00100100 11010110
8	57,065	11011110 11101001
9	20,960	01010001 11100000
10	10,347	00101000 01101011
11	3,450	00001101 01111010
12	44,861	10101111 00111101
13	58,916	11100110 00100100
14	45,023	10101111 11011111
15	61,022	11101110 01011110
16	6,865	00011010 11010001
17	23,720	01011100 10101000
18	46,227	10110100 10010011
19	11,138	00101011 10000010
20	13,733	00110101 10100101

27.1 The final seed value is 1043618065 The final random number is 0.4859725

Table 27.19: Chi-Square Test on 10,000 Numbers

Cell	Observed	Expected	$\frac{(\text{Observed}-\text{Expected})^2}{\text{Expected}}$
1	993	1000.0	0.049
2	1007	1000.0	0.049
3	998	1000.0	0.004
4	958	1000.0	1.764
5	1001	1000.0	0.001
6	1049	1000.0	2.401
7	989	1000.0	0.121
8	963	1000.0	1.369
9	1026	1000.0	0.676
10	1016	1000.0	0.256
Total	10000	10000.0	6.690

The computed statistic is 6.690. The 0.1-quantile of a chi-square variate with nine degrees of freedom is 4.168. The sequence doest not passes the test at 90%.

27.2 Fifteen random numbers generated using the given LCG are 6, 15, 12, 13, 2, 11, 8, 9, 14, 7, 4, 5, 10, 3, 0.

The normalized numbers obtained by dividing by 16 are 0.37500, 0.93750, 0.75000, 0.81250, 0.12500, 0.68750, 0.50000, 0.56250, 0.87500, 0.43750, 0.25000, 0.31250, 0.62500, 0.18750, 0.00000.

Table 27.20 shows a sorted list of these numbers and differences. Usin the maximum values obtained from the table, K-S statistics can be computed as follows:

$$K^+ = \sqrt{n}\max_{\mathrm{j}}\left(\frac{j}{n} - x_j\right) = \sqrt{15} \times 0.06667 = 0.2582$$

$$K^- = \sqrt{n}\max_{\mathrm{j}}\left(x_j - \frac{j-1}{n}\right) = \sqrt{15} \times 0.05833 = 0.2259$$

$K^+ = 0.2582$, $K^- = 0.2259$. Both values are less than $K_{[0.95;15]}$=1.1773. The sequence passes the test.

Table 27.20: Computation for the K-S Test

j	x_j	$\frac{j}{n} - x_j$	$x_j - \frac{j-1}{n}$
1	0.00000	0.06667	0.00000
2	0.12500	0.00833	0.05833
3	0.18750	0.01250	0.05417
4	0.25000	0.01667	0.05000
5	0.31250	0.02083	0.04583
6	0.37500	0.02500	0.04167
7	0.43750	0.02917	0.03750
8	0.50000	0.03333	0.03333
9	0.56250	0.03750	0.02917
10	0.62500	0.04167	0.02500
11	0.68750	0.04583	0.02083
12	0.75000	0.05000	0.01667
13	0.81250	0.05417	0.01250
14	0.87500	0.05833	0.00833
15	0.93750	0.06250	0.00417
	Max	0.06667	0.05833

Table 27.21: Autocovariances for the Random Sequence for exercise 27.3

Lag	Autocovariance	St. Dev.	90% Confidence Interval	
k	R_k	of R_k	Lower Limit	Upper Limit
1	-0.001696	0.000833	-0.002763	-0.000629 $\surd$
2	0.000429	0.000833	-0.000638	0.001495
3	-0.000039	0.000833	-0.001105	0.001028
4	-0.000221	0.000834	-0.001287	0.000846
5	0.001335	0.000834	0.000268	0.002402 $\surd$
6	-0.000420	0.000834	-0.001487	0.000647
7	0.000856	0.000834	-0.000211	0.001923
8	0.000630	0.000834	-0.000437	0.001698
9	-0.001105	0.000834	-0.002172	-0.000037 $\surd$
10	0.000964	0.000834	-0.000103	0.002032

27.3 The autocovariance and confidence intervals for the serial autocovariances at lags 1 to 10 are given in table 27.21. Three autocovariance (at lag 1, 5, 9) are significant.

27.4 The pairs generated by the first generator lie on the following two lines with a positive slope:

$$x_n = \frac{1}{2}x_{n-1} + \frac{13}{2}k, k = 0, 1$$

The distance between the lines is $13/\sqrt{5}$. The pairs generated by the second generator lie on the following two lines with a negative slope:

$$x_n = -2x_{n-1} + 13k, k = 1, 2$$

The distance between the lines is $13/\sqrt{5}$. Both generators have the same 2-distributivity.

28.1 a. Inverse transformation: Generate $u \sim U(0,1)$. If $u < 0.5$, then $x = \sqrt{2u}$; otherwise $x = 2 - \sqrt{2(1-u)}$.

b. Rejection: Generate $x \sim U(0,2)$ and $y \sim U(0,1)$. If $y \leq \min(x, 2-x)$, then output x; otherwise repeat with another pair.

c. Composition: The pdf $f(x)$ can be expressed as a weighted sum of a left triangular density and a right triangular density.

d. Convolution: Generate $u_1 \sim U(0,1)$ and $u_2 \sim U(0,1)$. Return $u_1 + u_2$.

29.1 a. Geometric. Geometric distribution is used to model number of attempts between successive failures.

b. Negative binomial. It can be used model the numberof failures before the mth success.

c. Logistic. (??) (xxx)

d. Normal. The mean of large set of uniform distribution is a normal distribution.

e. Lognormal. The product of large set of uniform is a lognormal distribution.

f. Pareto. This is used fit power curves.

g. Poisson. Sum of two Poisson's is a Poisson distribution.

h. Chi square. Variances of normal population has chi-square distribution.

i. F. Ratio of variances of normal population has F distribution.

j. Exponential. Exponential is used to model memoryless events.

k. Erlang-m. Sum m memoryless servers can be represented by Erlang-m distribution..

l. Binomial. This models the successes in n independent and identical Bernoulli trails.

m. Beta. This is used to model ratio of random-variates.

29.2 a. Sum of normal distribution is normal. $\mu = 0/4 = 0.\sigma = \sqrt{4/4} = 1$. Hence it is a $N(0, 1)$ distribution. 90% quantile is 1.281 from appendix table A.2

b. Sum of variances is chi-square distribution. There are 4 normal variates, hence it is a $\chi^2(4)$ distribution. 90% quantile is 7.779 from appendix A.5.

c. Ratio of two chi-square variates is a F distribution. Since both numerator and denominator have two variates the degrees of freedom is 2 for both. Hence it is a $F(2,\ 2)$ distribution. 90% quantile is 9.00 from appendix A.6.

d. Ratio of normal to square root of chi-square distribution is a t distinction. Number of degrees of freedom is 3. Hence it is a $t(3)$ distribution. 90% is 1.638 from appendix A.4.

30.1 Erlang-k arrivals, general bulk service, five servers, 300 waiting positions, 5000 population size, and last come first served preemptive resume service.

30.2 Because it provides 10 waiting positions for a population of only 5. Also, since there are 12 servers and only 10 waiting positions, two servers have no waiting positions.

30.3 Both will provide the same performance. Increasing buffers beyond the population size has no effect.

30.4 $E[n] = \lambda E[r] = \frac{10{,}800}{3600} \times \frac{1}{3} = 1$. Job flow balance was assumed. Service time distribution has no effect.

30.5 If $k = 1$ then E_k becomes a exponential distribution. Hence $\mathrm{E}_k/\mathrm{M}/1$ can be called a Poisson process if $k = 1$.

31.1 a. The probability p_n for birth-death processes is given by

$$p_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} p_0, n = 1, 2, \ldots, \infty$$

Here $\lambda_n = \frac{\lambda}{n+1}$ and $\mu_n = \mu$. Therefore

$$p_n = \frac{\rho^n}{n!} p_0, \text{where } \rho = \frac{\lambda}{\mu}$$

b. The sum of the probabilities should be 1. Therefore,

$$p_0 = \frac{1}{1 + \sum_{n=0}^{\infty} \frac{\rho^n}{n!}}$$

$$p_0 = e^{-\rho}$$

c.

$$E[n] = \sum_{n=1}^{\infty} n p_n = p_0 \sum_{n=1}^{\infty} n \frac{\rho^n}{n!}$$

$$E[n] = e^{-\rho} \rho e^{\rho} = \rho$$

d. $\lambda' = \frac{\lambda}{\rho}(1 - e^{-\rho})$

e. $E[r] = \frac{E[n]}{\lambda'} = \frac{\rho^2}{\lambda(1-e^{-\rho})}$

31.2 a. The probability p_n for birth-death processes is given by

$$p_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} p_0, n = 1, 2, \ldots, \infty$$

Here $\lambda_n = \lambda$ and $\mu_n = n\mu$. Therefore

$$p_n = \frac{\rho^n}{n!} p_0, \text{where } \rho = \frac{\lambda}{\mu}$$

b. $p_0 = e^{-\rho}$ (derivation is similar to previous exercise solution)

c. $E[n] = \rho$

d. $\text{Var}[n] = E[n^2] - (E[n])^2 = \sum_{n=1}^{\infty} n^2 \frac{\rho^n}{n!} p_0 - \rho^2 = e^{-\rho}\rho(\rho+1)e^{\rho} - \rho^2$
$\text{Var}[n] = \rho$

e. $E[r] = \frac{E[n]}{\lambda} = \frac{1}{\mu}$

31.3 a. $\rho = 50/60 = 5/6$
b. $E[s] = 1/\mu = E[r](1-\rho) = 3(1-5/6) = 3/6 = 0.5$ second
c. $\lambda = \mu\rho = 10/6 \Rightarrow 60(10/6) = 100$ queries per minute
d. $E[n] = \frac{\rho}{1-\rho} = \frac{5/6}{(1-5/6} = 5$
e. $P(n > 10) = \rho^{11} = (5/6)^1 1 = 0.135$
f. $r_{90} = E[r]\ln[10] = 6.9$ seconds
g. $w_{90} = E[r]\ln[10\rho] = 6.36$ seconds

31.4 a. $\rho = \frac{\lambda}{m\mu} = \frac{30}{3\times(1/0.05)} = 0.5$
b. $p_0 = 0.21$
c. $\varrho = \frac{(3\times 0.5)^3}{3!(1-0.5)} 0.21 = 0.24$
d. $E[n] = 3 * 0.5 + 0.5 \times 0.24/(1-0.5) = 1.74$
e. $E[n_q] = 0.5 \times 0.24/(1-0.5) = 0.24$
f. $E[r] = \frac{1}{20}\left(1 + \frac{0.24}{3(1-0.5)}\right) = 0.0579$ second
g. $\text{Var}[r] = 0.00296$ second^2 (use forumla 14 of Box 31.2)
h. $w_{90} = 0.0287$ second (use formula 19 of Box 31.2)

31.5 a. $\lambda = 30/3 = 10 \Rightarrow \rho = 10/(1/0.05) = 0.5$
b. $p_0 = \rho = 0.50$
c. $\varrho = 0.50$
d. $E[n] = \frac{0.5}{1-0.5} = 1$ request per drive
e. $E[n_q] = \frac{0.5^2}{1-0.5} = 0.5$ request per drive
f. $E[r] = \frac{1/20}{1-0.5} = 0.1$ second
g. $\text{Var}[r] = (E[r])^2 = 0.1 \times 0.1 = 0.01$ second^2
h. $w_{90} = 0.16$ second (use formula 20 of Box 31.1)

31.6 Yes. With the new system, the 90-percentile of the waiting time will be zero.

31.7 Yes, since with $\lambda = 0.167/2 = 0.0833$, average waiting time is 0.18 minutes and the 90-percentile of waiting time is zero.

31.8 a. $p_0 = 0.22$ (use formula 4 of Box 31.3)
$p_1 = 0.34, p_2 = 0.25, p_3 = 0.13, p_4 = 0.0629$ (use formula 5 of Box 31.3)
b. $E[n] = 1.5$ requests (use formula 6 of Box 31.3)
c. $E[n_q] = 0.0629$ requests (use formula 7 of Box 31.3)
d. $\text{Var}[n] = 1^2 \times p_1 + 2^2 \times p_2 + 3^2 \times p_3 + 4^2 \times p_4 - (1.5)^2 = 1.3$
e. $\lambda' = \lambda(1 - p_B) = 30(1 - 0.0629) = 28$ requests per second
f. Loss rate$= \lambda p_B = 30 \times 0.0629 = 1.9$ requests per second
g. $U = \rho(1 - p_B) = 0.5(1 - 0.0629) = 0.47$
h. $E[r] = E[n]/\lambda' = 1.5/28 = 0.0535$ second

31.9 The probability p_n for birth-death processes is given by

$$p_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} p_0, n = 1, 2, \ldots, \infty$$

$$p_n = \begin{cases} p_0 (m\rho)^n \begin{pmatrix} K \\ n \end{pmatrix} & 0 \le n < m \\ p_0 \rho^n \begin{pmatrix} K \\ n \end{pmatrix} \frac{n!}{m!} m^m & m \le n \le K \end{cases}$$

where $\rho = \frac{\lambda}{m\mu}$
Average throughput $\lambda' = \sum_{n=0}^{K-1} (K - n)\lambda p_n = \lambda(K - E[n])$
$U = \frac{\lambda'}{m\mu} = \rho(K - E[n])$
$E[r] = \frac{E[n]}{\lambda'} = \frac{E[n]}{\lambda(K - E[n])}$

31.10 $p_n = \begin{cases} \begin{pmatrix} K \\ n \end{pmatrix} (m\rho)^n p_0 & 0 \le n < m \\ \begin{pmatrix} K \\ n \end{pmatrix} \frac{n!(\rho)^n m^m}{m!} p_0 & m \le n \le B \end{cases}$

where, $\rho = \frac{\lambda}{m\mu}$.
Average throughput $\lambda' = \sum_{n=0}^{B} (K - n)\lambda p_n = \lambda \left(K - E[n] - (K - B)p_B\right)$
where p_B is the probability of B jobs in the system.
$U = \frac{\lambda'}{m\mu} = \rho \left(K - E[n] - (K - B)p_B\right)$
$E[r] = \frac{E[n]}{\lambda'} = \frac{E[n]}{\lambda(K - E[n] - (K - B)p_B)}$

32.1 a. Job flow balance. Number of job arrivals is not equal to number of departures since some jobs are lost.

b. Fair service

c. Single resource possession

d. Routing homogeneity

e. No blocking

f. Single resource possession

g. One-step behavior

33.1 X=400/10=40, S=1/200, $U = XS$=40/200=20%

33.2 n=4, X=5, $n = XR \Rightarrow R$=4/5 second.

33.3 X=40/10=4, V_{disk}=2, $X_{\text{disk}} = XV_{\text{disk}}$=8, S_{disk}=0.030, $U_{\text{disk}} = X_{\text{disk}}S_{\text{disk}}$=8 $\times$ 0.03 = 0.24 $\Rightarrow$ 24%

33.4 X_{printer}=500/10=50, V_{printer}=5, $X = X_{\text{printer}}/V_{\text{printer}}$= 50/5 = 10 jobs/minute

33.5 a. $V_{\text{CPU}} = 1/0.04 = 25$, $V_A = 0.8/0.04 = 20$, $V_B = 0.16/0.04 = 4$.
b. D_{CPU}=0.04 $\times$ 25 = 1, D_A=0.03 $\times$ 20 = 0.6, D_B= 0.025$\times$4 = 0.1
c. $U_k = XD_k \Rightarrow X$=0.6/0.6=1; U_{CPU}=1, U_B=0.1
d. $U_k = XD_k \Rightarrow X$=0.1/0.1=1; $R = N/X - Z = 20/1 - 5 = 15$ seconds

33.6 (xxx The data used here looks like that from problem 34.1) a. CPU (since it has the most demand)
b. $R_{\text{min}} = D_1 + D_2 + D_3$ = 1+0.6+0.1 = 1.7
c. X $\leq 1 \Rightarrow U_2 = 1 \times 0.6 = 60\%$
d. D_{max}=1, $U_k = XD_k \Rightarrow X \leq 1$ job/second
e. $R \geq \max\{D, ND_{\text{max}} - Z\} \Rightarrow D_{\text{max}} \leq \frac{R+Z}{N} = 0.6$. We need at least a 40% faster CPU; disk A would be just OK.
f. D=1.7, D_{max}=1, Z=5, $\Rightarrow X \leq \min\{\frac{N}{6.7}, 1\}$, $R \geq \max\{1.7, N - 5\}$

33.7 a. D_1=0.5,D_2 = 0.6,D_3 = 0.1 $\Rightarrow$ disk A
b. D_1=0.1, D_2=1.2,D_3 = 0.1 $\Rightarrow$ disk A
c. D_1=1,D_2 = 0.6,D_3=0.2 $\Rightarrow$ CPU
d. D_1=1,D_2 = 0.6,D_3=2 $\Rightarrow$ disk B

34.1 $X = 0.8$ $D_1 = 1$, $D_2 = 20 \times 0.03 = 0.6$, $D_3 = 4 \times 0.25 = 0.1$, $Q_i = U_i/(1-U_i) = XD_i/(1-XD_i)$, $Q_1 = 0.8 \times 1/(1-0.8 \times 1) = 4$, $Q_2 = 0.8 \times 0.6/(1-0.48) = 0.923$, $Q_3 = 0.8 \times 0.1/(1-0.08) = 0.087$ $Q_avg = \sum_i Q_i = 4 + 0.923 + 0.087 = 5.01$.
$R_i = S_i/(1-U_i) = S_i/(1-XD_i)$, $R_1 = 0.04/(1-0.8) = 0.2$, $R_2 = 0.03/(1-0.48) = 0.0577$, $R_3 = 0.025/(1-0.08) = 0.0272$, $R_avg = \sum R_i V_i = 0.2 \times 25 + 0.0577 \times 20 + 0.0272 \times 4 = 6.2628$ seconds.

34.2 (xxx depends on data of 33.6 which is wrong!)

	Response Time				System	Queue Lengths		
N	CPU	Disk A	Disk B	System	Throughput	CPU	Disk A	Disk B
1	0.040	0.030	0.025	1.700	0.149	0.149	0.090	0.015
2	0.046	0.033	0.025	1.904	0.290	0.333	0.189	0.029
3	0.053	0.036	0.026	2.149	0.420	0.559	0.299	0.043
4	0.062	0.039	0.026	2.443	0.537	0.838	0.419	0.056
5	0.074	0.043	0.026	2.795	0.641	1.179	0.546	0.068

34.3 (xxx depends on data of 33.6 which is wrong!)

Itera-	Response Time				System	Queue Lengths		
tion No.	CPU	Disk A	Disk B	System	Throughput	CPU	Disk A	Disk B
1	0.293	0.220	0.183	12.467	1.145	8.397	5.038	0.840
2	0.359	0.174	0.045	12.629	1.135	10.185	3.939	0.204
3	0.427	0.142	0.030	13.640	1.073	11.454	3.053	0.128
4	0.475	0.117	0.028	14.334	1.034	12.291	2.421	0.116
5	0.507	0.099	0.028	14.767	1.012	12.826	2.003	0.112

34.4 Each packet is serviced by the 3 computers. Hence the throughput is when there is one packet is $X_1 = 1/3S$. The packet spents a time of S in each hop so the response time is $R_1 = 3S$. Similiarily
$X_2 = 2/4S$, $R_2 = 4S$
$X_3 = 3/5S$, $R_3 = 5S$
$X_4 = 4/6S$, $R_4 = 6S$
$X_5 = 5/7S$, $R_4 = 7S$
In general:
$R = (n+2)S$, $X = \frac{n}{(n+2)S}$, $n \geq 1$

34.5 If there are h hops in the network, then the each packet has to get serviced by h hops and has to wait a time of nS hops before getting serviced. Hence,

the response time is $R = (n+h)S$. Arguing similarily the throughput is $X = \frac{n}{(n+h)S}$. Power $X/R = \frac{n}{(n+h)^2S^2}$ is maximum when $(n+h)^2 - 2(n+h)n = 0 \Rightarrow n = h$

34.6 (xxx depends on data of 33.6 which is wrong!)
The balanced job bounds are

$$\frac{N}{5+1.7+(N-1)0.57\frac{1.7(N-1)}{1.7(N-1)+5}} \le X(N) \le \min\left\{1, \frac{N}{5+1.7+(N-1)\frac{1.7}{1.7+5}}\right\}$$

$$\max\left\{N-5, 1.7+(N-1)\frac{1.7}{1.7+5}\right\} \ge R(N) \le 1.7+(N-1)0.57\frac{(N-1)1.7}{(N-1)1.7+5}$$

	Response Time			Throughput		
	Lower		Upper	Lower		Upper
N	BJB	MVA	BJB	BJB	MVA	BJB
1	1.700	1.700	1.700	0.149	0.149	0.149
2	1.844	1.904	1.954	0.288	0.290	0.292
3	1.988	2.149	2.510	0.399	0.420	0.429
4	2.131	2.443	3.215	0.487	0.537	0.561
5	2.275	2.795	4.005	0.555	0.641	0.687
6	2.419	3.213	4.848	0.609	0.731	0.809
7	2.563	3.706	5.726	0.653	0.804	0.926
8	3.000	4.278	6.629	0.688	0.862	1.000
9	4.000	4.930	7.549	0.717	0.906	1.000
10	5.000	5.658	8.483	0.742	0.938	1.000

34.7 a. $X = \frac{N}{D(1+\frac{N-1}{M})} = \frac{NM}{D(M+N-1)}$, $R = Q/X = N/X = \frac{D(M+N-1)}{M}$

b. Substituting $D_{\text{avg}} = D_{\max} = D/M$ and $Z = 0$ in Equations (34.8) and (34.9), we get the same expressions as in Exercise 34.6 for balanced job bounds.

Table 35.22: Computing the Normalizing Constant for Exercise 35.1

n	$y_{\mathrm{CPU}} = 10$	$y_A = 6$	$y_B = 1$
0	1	1	1
1	10	16	17
2	100	196	197
3	1000	2176	2177

35.1 $D_{\mathrm{CPU}} = 0.04 \times 25 = 1$, $D_A = 0.03 \times 20 = 0.6$, $D_B = 0.025 \times 4 = 0.1$. For scaling factor choose $\alpha = 1/0.1 = 10$. This results in $\mathrm{y}_{\mathrm{CPU}} = 10$, $\mathrm{y}_A = 6$, y_B=2.

The probability of exactly j jobs at the ith device is:

$$\begin{aligned} P(n_i = j) &= P(n_i \geq j) - P(n_i \geq j+1) \\ &= \frac{y_i^j}{G(N)} \left(G(N-j) - y_i G(N-j-1) \right) \end{aligned}$$

$$P(Q_{\mathrm{CPU}} = 0 | N = 3) = \frac{1}{2177}(2177 - 10 \times 197) = 0.095$$

$$P(Q_{\mathrm{CPU}} = 1 | N = 3) = \frac{10}{2177}(197 - 10 \times 17) = 0.124$$

$$P(Q_{\mathrm{CPU}} = 2 | N = 3) = \frac{100}{2177}(17 - 10 \times 1) = 0.322$$

$$P(Q_{\mathrm{CPU}} = 3 | N = 3) = \frac{1000}{2177}(1 - 10 \times 0) = 0.459$$

(xxx the book answer is different. $P(Q_{\mathrm{CPU}} = n | N = 3)$ for $n = 0$, 1, 2, 3 are 0.108, 0.180, 0.293, and 0.419, respectively.).

The throughputs for $N = 1, 2, 3$ are

$$X(3) = \alpha \frac{G(N-1)}{G(N)} = 10 \times 197/2177 = 0.905$$

$$X(2) = 10 \frac{G(1)}{G(2)} = 10 \times 17/197 = 0.863$$

$$X(1) = 10 \frac{G(0)}{G(1)} = 10 \times 1/17 = 0.588$$

(xxx in book are different)

35.2 (xxx data in 33.6 is wrong!) $P(Q_{\mathrm{CPU}} = n|N = 3)$ for $n = 0$, 1, 2, 3 are 0.580, 0.298, 0.103, and 0.018, respectively.

36.1 (xxx answer depends on data of Exercise 35.1) $X = 0.588$, 0.796, and 0.892 for $N = 1$, 2, 3, respectively; $R = 1.700$, 2.506, 3.365 for $N = 1$, 2, 3, respectively.

36.2 (xxx answer depends on data of Exercise 35.1) The service rates of the FEC are 1.429, 1.628, and 1.660, respectively, for one through three jobs at the service center.